The template channel dominates, the MSA selects, and sequence barely matters: family-consensus conformational-state selection in an AlphaFold3-style protein structure network

Cheng Ding[1,*]

[1]School of Life Science and Technology, ShanghaiTech University, Shanghai, China

[*]Correspondence: dingcheng2024@shanghaitech.edu.cn; ORCID: 0009-0001-7221-9066

Abstract

Structure predictors of the AlphaFold family receive three conceptually distinct inputs — the query sequence, a multiple sequence alignment (MSA), and structural templates — and a substantial body of work on AlphaFold2 (AF2) has established a functional hierarchy among them: a strong MSA overrides templates, and both dominate the bare sequence. Whether this hierarchy survives the architectural redesign of AlphaFold3-style (AF3-style) trunks — in which the MSA is demoted to a single small module that writes once into the pair representation, and the template track is trained directly against noised ground-truth structures — is unknown. We address this question in RosettaFold-3 (RF3), an open AF3-style model, using a distogram-based conformational-state preference score that asks, for 13 proteins with two experimentally resolved states, which state the network favours under each input condition. Three results emerge. First, the MSA acts as a *family-consensus state selector*, not a generic crystallographic-bias agent: it shifts RF3 away from the crystallized state in KaiB and RfaH (shifts −2.29 and −2.99 in NLL units), toward it in XCL1 and Mad2 (+2.87, +0.91), sharpens the sequence-intrinsic fold preference of the designed GA88/GB88 pair (+2.56), and leaves a GPCR, two kinases, a transporter and three X-ray-versus-NMR control pairs inside the measured noise band. Second, the hierarchy of the input channels is *inverted* relative to AF2: experimental templates steer state preference by ±4–5 units and override the MSA in direct conflict on all tested targets (template ±4–5 > MSA ±2.7 > sequence ±0.3 on GA88/GB88), while on a deep mutational scanning benchmark a wild-type crystal template is actively *harmful* (−0.049 Spearman relative to no-MSA) where the MSA is helpful — the MSA is not a de facto template. Third, in a homodimer the MSA preferentially amplifies the inter-chain block of the pair representation, but an unpaired control shows this is amplification of a network-internal interface prior by family context, not inter-chain coevolution. All conclusions are model-specific to the AF3-style template channel and should not be extrapolated to AF2-class architectures without retesting.

1 Introduction

A non-trivial minority of proteins adopt more than one stable, experimentally characterizable conformation: metamorphic fold-switchers such as KaiB, RfaH, Mad2 and lymphotactin (XCL1) refold secondary structure wholesale [25, 27, 28, 29, 30], while GPCRs, kinases and transporters populate alternative functional states that differ more modestly but consequentially. Deep-learning structure predictors, trained on the Protein Data Bank and evaluated largely on single-state accuracy, compress this plurality into one answer: AlphaFold2 (AF2) captures only one of the two known conformers for the large majority of fold switchers [16, 15], and the AF3 authors themselves note that their model "typically predict[s] static structures as seen in the PDB" [2]. Understanding *which input channel* decides the predicted state is therefore both a mechanistic question about these networks and a practical one for anyone trying to sample alternative conformations.

For AF2 the answer is well documented. The Evoformer couples a processed MSA to the pair representation over many blocks [1], templates enter as a separate and comparatively weak track, and when the two conflict, the MSA wins: "if the MSA signal is strong, AlphaFold tends to ignore structural information from the template" [4]. An entire methodological family exploits this hierarchy — subsampling, clustering, mutating or masking the MSA releases alternative conformational states [10, 11, 12, 13] — and removing the MSA removes learned state biases in GPCR modelling [19, 18]. The accepted ordering is thus MSA > template > sequence for conformational control in AF2-class models.

The MSA itself entered structure prediction as the carrier of evolutionary covariation: residue pairs that co-vary across a family tend to be in contact, and coevolution-only methods could fold small proteins without any structural training data [9]. AlphaFold1 distilled these statistics into learned distance potentials [8]; AF2 made the MSA a first-class, actively processed input. The field has since moved in both directions at once. Single-sequence language models removed the MSA while retaining most of its structure signal [5, 6], and generative design models carry the structural prior entirely in pretrained weights [7]. Yet both AF2 and AF3 retain a measurable dependence on MSA depth [1, 2], and alignment-based statistics remain among the strongest substrates for variant-effect prediction across MSA depths [21, 22]. The MSA is neither dispensable nor neutral: it is a channel with real information content whose conformational side effects are the subject of this paper.

AF3-style trunks reorganize these channels in ways that make the AF2 answer non-obvious. The MSA is "substantially de-emphasized": four small blocks integrate it by pair-weighted averaging, after which the MSA representation is discarded and all information flows through the pair representation [2]. RosettaFold-3 (RF3), the open implementation we study, follows this design [3]. In parallel, the template input was upgraded to an atom-level track trained against noised ground-truth structures [3] — a training regime that could plausibly teach near-literal copying, in contrast to AF2's heavily regularized template path. Two channels with opposite directions of architectural travel (demoted MSA, empowered template) motivate a direct re-measurement of the hierarchy, and of what the MSA's conformational influence actually consists in.

Existing evidence about MSA effects is almost entirely input-to-output: perturb the MSA, observe structures or fitness correlations [10, 11, 12, 17]. Two gaps matter here. First, no study has quantified, inside an AF3-style trunk, how the MSA and a template each move the network's conformational preference on proteins with two known states. Second, the natural suspicion — that an MSA acts as a distributed, evolution-averaged template and that "MSA bias" is just crystallographic bias in disguise [19, 18, 4] — has never been tested by pitting a real template against a real MSA in the same network, at the level of output state preference, internal representation geometry, and downstream function.

We fill both gaps in RF3. On 13 two-state targets scored by distogram negative log-likelihood (NLL) against both experimental reference structures, we show that the MSA selects a family-consensus state with target-dependent direction, that a ground-truth-trained template channel dominates the MSA in direct conflict (inverting the AF2 hierarchy), that template and MSA writes are qualitatively different objects (the template is a stronger, mutation-erasing rewrite that harms variant-effect prediction, the MSA a milder, family-consensus write that helps), and that the MSA's amplification of inter-chain geometry in a homodimer reflects an internal interface prior rather than paired-alignment coevolution. Throughout, we report only quantities computed on shared residue universes after correcting two analysis bugs (a span-mismatch/compression bug and a silently dropped template input), and we flag single-forward noise limits explicitly.

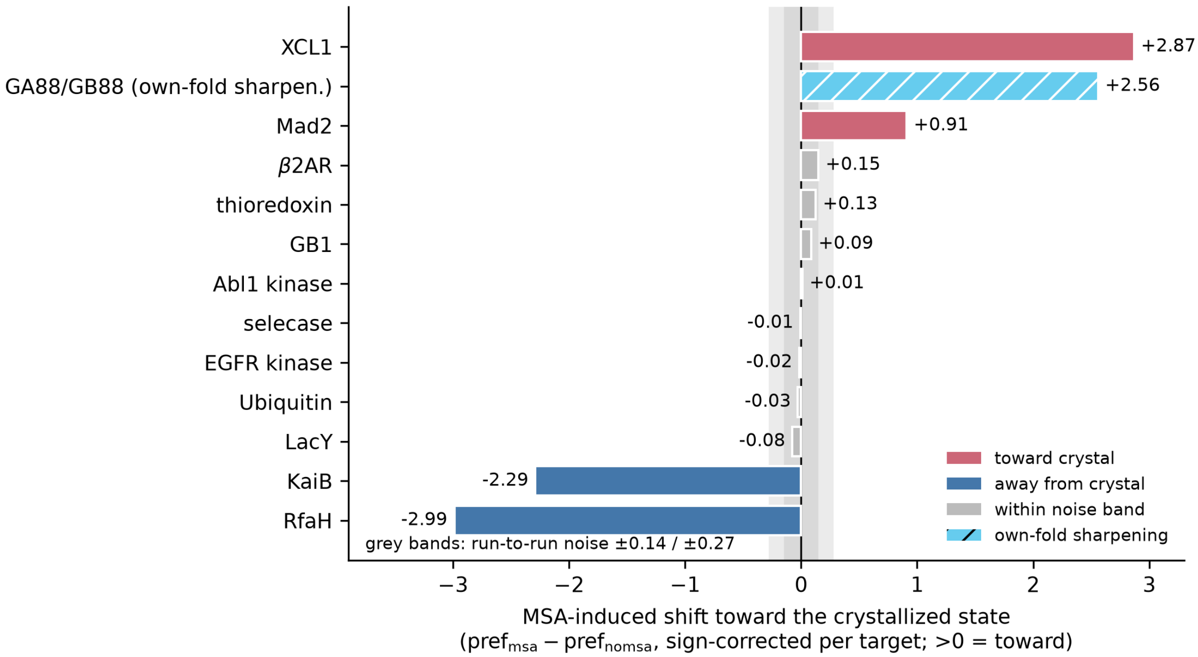

Figure 1: The MSA shifts conformational state in a target-dependent direction. MSA-induced shift of the distogram state preference, sign corrected so that positive values point toward the state that dominates the PDB for that target (for Mad2 the crystallized state is state B; the GA88/GB88 entry is an own-fold sharpening score and has no crystallographic direction). Shaded bands: run-to-run preference noise (±0.14 inner, ±0.27 outer). Red: significant toward-crystal; blue: significant away-from-crystal; grey: within noise. All values from the range-matched rescoring (Table 1).

2 Results

2.1 A family-consensus state selector, not a crystal-bias agent

We assembled 13 proteins with two experimentally resolved states each, spanning fold-switchers (KaiB, RfaH, Mad2, XCL1, selecase, the designed GA88/GB88 pair), a GPCR (β2AR), two kinases (Abl1, EGFR), a transporter (LacY), and three same-construct X-ray-versus-NMR pairs (ubiquitin, GB1, thioredoxin) that serve as near-null controls (Table 1). For each target we ran RF3 with and without its MSA and scored the predicted distogram against both experimental Cα distance maps on a shared residue universe, defining pref $= \mathrm{NLL}_B - \mathrm{NLL}_A$ (positive favours state A) and the MSA-induced shift Δpref $= \mathrm{pref}_{\mathrm{msa}} - \mathrm{pref}_{\mathrm{nomsa}}$. Repeat forwards of identical inputs place the run-to-run noise band of Δpref at roughly ±0.14–0.27; we treat shifts inside this band as null (Methods).

The result is not a uniform pull toward crystallized states (Fig. 1). Two fold-switchers are shifted *away* from the crystallized ground state toward the alternative, NMR-characterized fold: KaiB (Δpref $= -2.29$; the MSA reverses the sign of the preference, $+0.72 \to -1.57$) and RfaH (-2.99; $+0.10 \to -2.89$). Two others are shifted *toward* the crystallized state: XCL1 ($+2.87$; $-0.78 \to +2.09$) and Mad2 (-0.91 with state B crystallized; $-0.05 \to -0.96$). The designed 88%-identical pair GA88/GB88 shows a third behaviour: RF3 already predicts each sequence into its own fold from sequence alone (GA88 $+0.38$; GB88 -0.23), and the MSA *sharpens* this intrinsic preference roughly tenfold ($+2.74$ and -2.66 respectively; $+2.56$ when GA88 is scored in the main range-matched flow) rather than redirecting it. Finally, the membrane and control targets sit at or inside the noise band: β2AR +0.15, thioredoxin +0.13, GB1 +0.09, Abl1 +0.01, selecase −0.01, EGFR −0.02, ubiquitin −0.03, LacY −0.08. We note explicitly that LacY and EGFR, which appeared to shift toward the crystallized state in a preliminary analysis (+0.94, +0.31), collapse to null once both states are scored on a shared residue universe; the "transporter/kinase/GPCR pulled toward the PDB state" pattern does not survive range matching, and we do not claim it.

The organizing principle that fits all four classes is that the MSA writes a *family-consensus* state. Where the family consensus coincides with the crystallized state (XCL1, Mad2), the shift is toward crystal; where the consensus favours the alternative fold (the KaiB and RfaH families, in which the fold-switched form is the more common family member [27, 28]), the shift is away from crystal. Crystal bias is thus a special case of consensus selection, not its definition — echoing output-level analyses of AF2 that attributed conformational pinning to evolutionary consensus averaging rather than to the PDB distribution per se [17]. For near-identical designed pairs, the family level is nearly uninformative and the MSA instead amplifies the sequence-encoded preference, consistent with the observation that MSA subsampling fails to release alternative states for some targets in AF2 as well [11].

Two technical points deserve emphasis. First, the tally is: 2 targets significantly toward crystal (XCL1, Mad2), 2 significantly away (KaiB, RfaH), 1 sharpening (GA88/GB88), and 8 null — so on a simple count the MSA is *inert* on the majority of our panel, and the secure signal lives entirely in the fold-switching targets. Second, the controls behave as controls should: the three X-ray-versus-NMR same-construct pairs and the single-homolog selecase MSA give $|\text{shift}| \leq 0.13$, so the distogram readout does not manufacture crystal preference where the two reference states differ only by experimental method, and a content-free MSA produces no shift.

2.2 The hierarchy inverts: template > MSA > sequence

For the five targets with usable structures of *both* states (KaiB, RfaH, Mad2, XCL1, LacY), we additionally supplied each state structure as a token-level template, alone (tplA, tplB) and in direct conflict with the MSA (msa_tplA: the MSA, whose direction was measured above, plus the template of the state the MSA pushes *away* from on KaiB/RfaH; the construction is analogous on all five).

Three observations stand out (Fig. 2, Table 2). First, templates steer strongly and bidirectionally: the template-steering span $\text{pref}_{\text{tplB}} - \text{pref}_{\text{tplA}}$ reaches −6.59 on KaiB (tplA +3.59 vs tplB −3.00) and is directionally template-following on all five targets (RfaH −1.19, Mad2 −1.70, LacY −0.47, XCL1 +0.63). Second, in direct conflict the template wins everywhere: the msa_tplA preference equals the tplA preference within $|\Delta| \leq 0.23$ on all five targets (e.g. KaiB: MSA alone −1.57 vs tplA +3.59; combined +3.36). The MSA's conformational influence is real but defeasible; the template's is effectively absolute at this distogram readout. This inverts the documented AF2 behaviour, in which a strong MSA overrides templates [4]. Third, the two channels' biases are distinct in kind: templates copy whichever state is supplied — crystal or NMR, with no intrinsic crystallographic preference — while the MSA writes family consensus.

The designed pair makes the ordering quantitative (Fig. 3). Sequence alone sets only a weak own-fold preference (GA88 +0.38, GB88 −0.23). The MSA amplifies it to ±2.7. A template of the protein's *own* fold strengthens it further (+4.88, −4.34), and a template of the *opposite* fold reverses it outright (−4.30, +4.75) — even in combination with the sequence's own MSA (−4.24, +4.58). In RF3 the conformational state-selection hierarchy is therefore template (±4–5) > MSA (±2.7) > sequence (±0.2–0.4). This contrasts with AF2, where memorized folds resist alternative templates [15] and strong MSAs override template information [4]; we return to the likely mechanistic reason in the Discussion.

One target requires an honest caveat. For Mad2, the state-A template (the O-state NMR

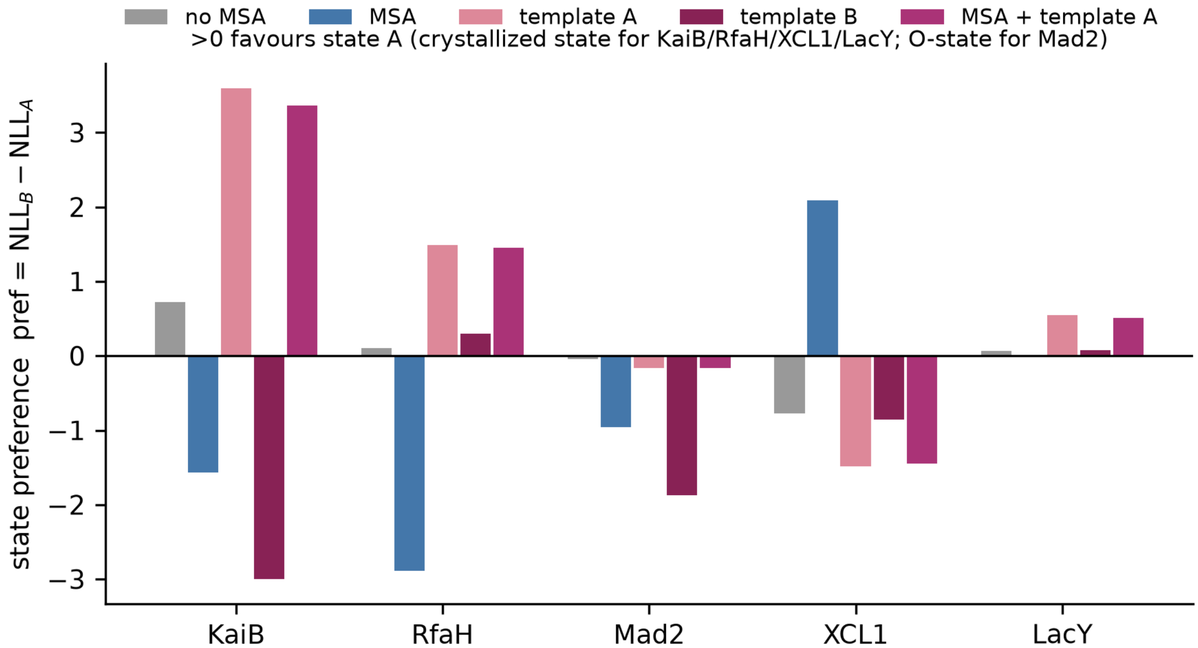

Figure 2: Templates steer strongly and bidirectionally; the MSA never overrides a conflicting template. State preference (pref = $\mathrm{NLL}_B - \mathrm{NLL}_A$; positive favours state A) for five two-state targets under sequence-only, MSA, template-of-state-A, template-of-state-B, and MSA-plus-template-A conditions. All values are range-matched. Note the Mad2 anomaly: the O-state template (1DUJ, NMR) fails to pull toward state A, while the C-state template works; see text.

structure 1DUJ) barely pulls toward A (tplA -0.17 vs sequence-only -0.05), while the state-B template works as expected (-1.87). 1DUJ is an N-terminal truncation and an NMR ensemble rather than a crystal structure; construct mismatch or ensemble averaging plausibly blunt its template effect. We report the anomaly rather than smoothing over it, and note that the conflict conclusion above does not depend on this arm.

2.3 A crystal template is not an MSA: three levels of falsification, and opposite functional signs

If the MSA's effect were equivalent to supplying a crystallographic template, the two should look alike wherever we look. They do not. We compared, on the SARS-CoV-2 spike receptor-binding domain (RBD), the pair representations and the downstream predictability produced by the wild-type crystal template 6M0J [33] versus a 15-row MSA, against the no-MSA reference (Fig. 4, Table 3).

At the **output-preference** level (Section above), the template dominates the MSA in conflict; the MSA does not mimic the template. At the **representation** level, both priors compress the effective rank of the pair representation (MSA 181 $\to$ 139, template 181 $\to$ 160), but the writes differ qualitatively: the template rotates the representation further from the no-MSA reference (CKA 0.443 vs 0.568), scrambles per-mutant delta directions harder (median cosine to the no-MSA delta 0.26 vs 0.48), and amplifies mutation-delta norms more (1.51 vs 1.21). The two arms themselves are only moderately similar (CKA(template, MSA) = 0.482), and the combined arm is indistinguishable from the template alone on every metric ($|\Delta| \leq 0.01$) — the template dominates geometrically as

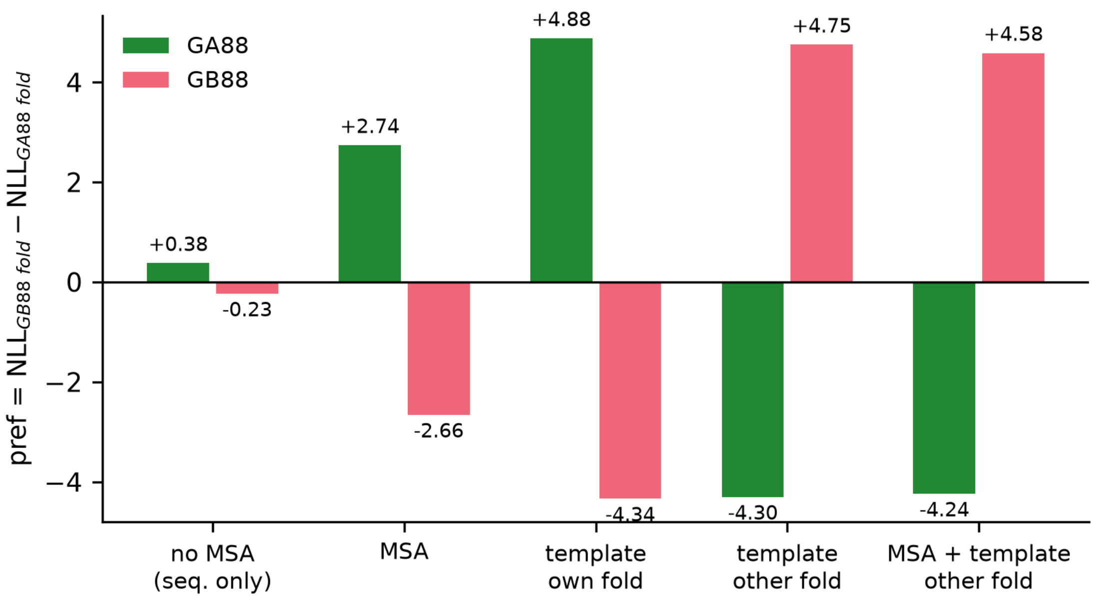

Figure 3: An opposite-fold template flips the designed GA88/GB88 pair even against sequence and MSA. Preference (positive favours the GA88 3α-helix fold) for both sequences under five input conditions. All four nomsa/msa values and both template arms are shown; the arms are nearly mirror-symmetric between the two 88%-identical sequences.

well. At the **functional** level, the contrast is sharpest: predicting per-position mutation effects from the pair representation (RBD DMS, 500-mutant subset, 5 splits $\times$ 3 initializations, cross-position Spearman ρ), the 15-row MSA improves over no-MSA (0.638 vs 0.415, +0.223 in this protocol; a de-biased full-set estimate of the MSA effect is +0.14–0.15 *in the same direction*), while the crystal template is *worse than supplying no evolutionary information at all* (0.366, -0.049 relative to no-MSA), and adding the MSA to the template does not rescue it (0.370).

The mechanistic reading is that the template's write is mutation-erasing: because the template track is trained on ground-truth structures, it pulls every mutant's pair representation toward the wild-type crystal, collapsing exactly the variation a DMS predictor needs. "Prior strength" and "content type" are independent axes: the template is the stronger prior yet carries the wrong content for variant analysis, while the MSA's milder write carries family-consensus content that is directly useful. Any temptation to think of the MSA as "a crystallographic template in disguise" — a natural reading of the AF2-era state-bias literature [19, 18] — is falsified at all three levels in this trunk.

2.4 Inter-chain amplification is an internal interface prior, not paired-alignment coevolution

Finally we asked whether the MSA's influence includes inter-chain coevolution, using human caspase-3 (CASP3), an obligate homodimer modelled as a 488-token two-chain input [32]. Across 392 mutants, the MSA restructures the within-chain blocks of the pair representation only moderately (median cosine between MSA and no-MSA blocks 0.712 for A–A and 0.712 for B–B) while the inter-chain A–B block is both better preserved in direction (0.749; A–B minus A–A = +0.037, paired Wilcoxon $p = 5.5 \times 10^{-66}$) and markedly amplified in norm (ratio 1.28 vs 1.05 within-chain). This is the signature of the network "docking" the two chains more emphatically when family

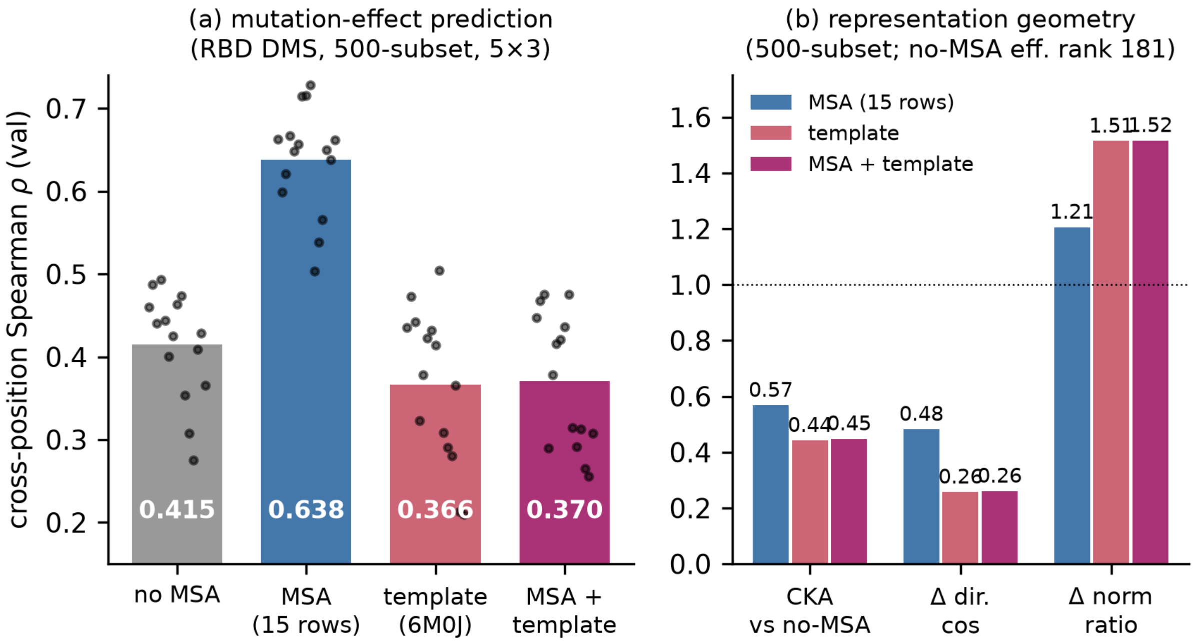

Figure 4: Template and MSA writes differ geometrically and diverge in function. (a) Cross-position Spearman correlation of predicted versus measured mutation effects on the RBD (dots: 15 split×init runs; bar: mean). The MSA helps; the wild-type crystal template harms. (b) Pair representation geometry relative to the no-MSA arm: CKA similarity, median cosine of mutation-delta directions, median mutation-delta norm ratio. The template arm is a stronger, more disruptive rewrite, and the combined arm tracks the template.

context is present.

To test whether this requires true inter-chain coevolution — homolog rows paired by species across the two chains — we repeated the extraction with the chain-B alignment rows shuffled to destroy row-wise pairing. The block statistics are identical to the paired run to three decimal places (Fig. 5: A–A 0.713/1.055, B–B 0.712/1.054, A–B 0.749/1.280; A–B minus A–A +0.037, $p = 3.9 \times 10^{-18}$; $n = 100$ mutants), even though per-mutant embeddings differ between the two MSAs at cosine 0.969 (the level expected from alignment subsampling noise). Amplifying the interface therefore does not require pairing information: the MSA's family context switches on (or turns up) an interface prior already present in the trunk. This is consistent with reports that paired alignments are often unnecessary for complex prediction [20, 5], and it sharpens them: at the representation level the effect of the MSA on the interface is not a coevolutionary docking signal at all.

3 Discussion

Our results support three claims, all scoped to AF3-style trunks as implemented in RF3. First, the MSA's conformational influence is a family-consensus state selection: its direction is set by which state the family statistics favour, which may be the crystallized state (XCL1, Mad2), the alternative state (KaiB, RfaH), or merely a sharpening of the sequence-intrinsic preference (GA88/GB88), and is absent on targets whose family is uninformative or whose states differ subtly. Second, the input hierarchy of AF2 is inverted: a ground-truth-trained template channel steers state preference by

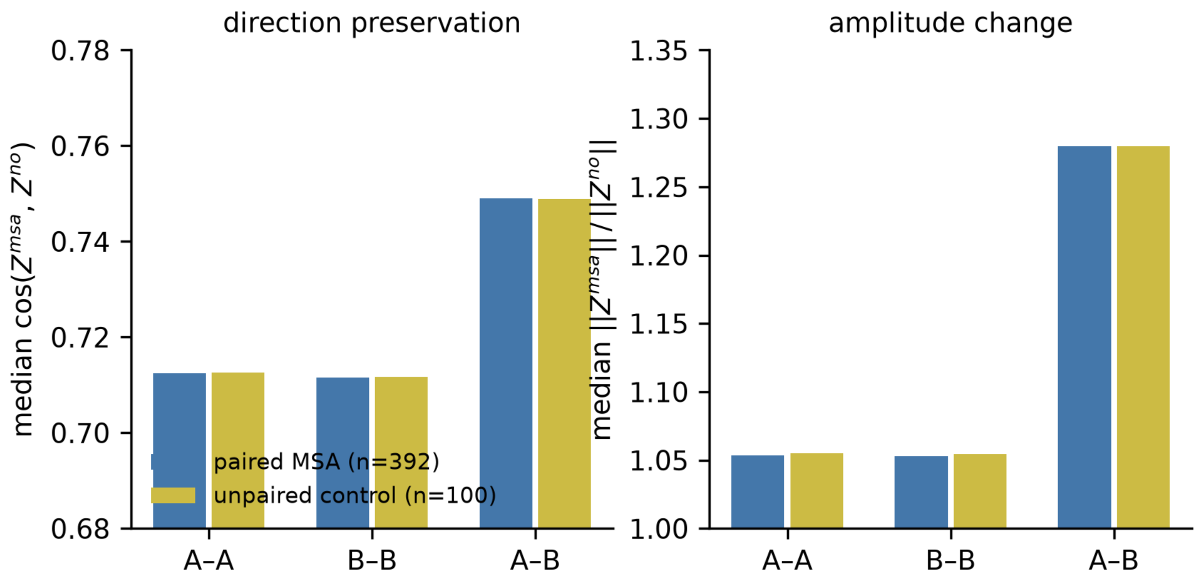

Figure 5: Inter-chain amplification survives destruction of row pairing. Block decomposition of the MSA effect on the CASP3 homodimer pair representation for the paired MSA (blue) and an unpaired control in which chain-B homolog rows were shuffled (yellow). Left: direction preservation (median cosine with the no-MSA block). Right: amplitude change (median norm ratio). The cross-chain block is preferentially amplified in both runs, identically.

±4–5 NLL units, dominates the MSA in direct conflict on every target tested, and flips even a designed fold pair against its own MSA, while the sequence alone contributes only ±0.2–0.4. Third, "the MSA as a de facto crystal template" fails on three independent levels, and the sign of the functional effect separates the two channels: family-consensus content helps mutation-effect prediction, a wild-type template hurts it.

Why might the hierarchy be inverted? The most parsimonious explanation is architectural. AF2 regularized its template track and iterated MSA–pair exchange over 48 Evoformer blocks; its templates were additionally downweighted by design choices such as template dropout during training, and empirically a strong MSA overrides templates [1, 4]. AF3-style models demote the MSA to a one-shot, pair-weighted write and simultaneously train the template track against noised ground-truth atom coordinates [2, 3], a regime that plausibly instils a literal-copying prior: the network has been rewarded, during training, for reproducing supplied structure. Under that prior, a template should behave like a near-irresistible instruction, which is what we observe. The same mechanism explains why the template write is mutation-erasing for DMS prediction: the copying prior has no notion of the query differing from the template.

Model specificity. This caveat deserves emphasis. Every number in this paper comes from one model (RF3), one checkpoint, and one inference configuration. AF2 and AF3 regularize their template channels differently, and other AF3-style implementations may place the template–MSA balance elsewhere. We therefore present the inverted hierarchy as a property of RF3's ground-truth-trained template channel and a demonstration that the AF2 ordering is not a law of the architecture family — not as a universal claim about all MSA-using predictors. The cleanest cross-model test would repeat the conflict arms of Fig. 2 and Fig. 3 in AF2 and AF3 proper.

Practical implications. For practitioners sampling alternative conformations, the AF2-era playbook — weaken or recluster the MSA to release alternative states [10, 11, 12, 13] — may be less effective, and less necessary, in AF3-style models: in RF3 the state is chosen primarily by the template track, so supplying a template of the desired state is the direct lever, and MSA manipulation becomes the secondary one. Conversely, users of AF3-style models for variant-effect or stability analysis should be aware that structural templates can actively degrade mutation-conditioned signal, while the MSA helps; and that MSA-driven interface amplification in homomers does not by itself evidence inter-chain coevolution.

Relation to the wider literature. Our measurements sit between two established observations. On one side, the claim that AF2-class networks learn an MSA–structure relationship rather than a sequence–structure relationship [24, 23] is borne out here in a precise sense: sequence alone barely moves the state preference ($\pm$0.2–0.4), while family context moves it by up to $\pm$3. On the other side, the MSA-perturbation literature shows that conformational diversity can also be released by network stochasticity alone, without touching the MSA [14], so input channels are not the only lever on state choice — and, in RF3, our conflict experiments show they are not even the strongest one. We also note a qualification our panel adds to the fold-switching literature: AF2's reported failure on fold switchers is partly memorization [15]; RF3, tested on a partly overlapping target set, gets the designed GA88/GB88 pair right from sequence alone, so fold-switch difficulty is itself model-dependent.

Limitations. (i) Each conformational preference rests on a single forward pass per condition; our noise band ($\pm$0.14–0.27) was estimated from repeat runs, and all headline shifts exceed it by at least $3\times$, but seed replicates would tighten the small effects we call null. (ii) The NLL preference is a distogram-level readout over shared residue universes, not a structure-level RMSD classification; it is sensitive and template-artifact-free, but it measures preference, not sampled ensembles. (iii) The two-state framing is a simplification — several targets populate more than two states. (iv) The Mad2 O-state template anomaly is unresolved; we favour a construct/ensemble explanation but cannot exclude a deeper asymmetry in the template channel. (v) The RBD functional comparison uses a single (if standard) DMS dataset and a shallow 15-row MSA; the sign contrast is the claim, not the exact effect sizes. (vi) Our conclusions draw on a corrected analysis pipeline: a first-pass scoring bug (states scored on non-shared residue spans, plus a 14-residue misalignment from a compressed disordered loop in the RfaH reference) inflated some shifts and manufactured others (LacY, EGFR), and an early template run silently dropped the template input; we report only post-correction values and regard the discarded analyses as cautionary, not evidentiary.

4 Methods

4.1 Model and inference

All experiments use RosettaFold-3 [3], an open implementation of the AF3 architecture, with a single checkpoint and inference settings held fixed throughout: 10 recycles, diffusion sampling with 50 steps and diffusion batch size 2. Per condition we record the distogram head logits (64 distance bins over 2–22 Å plus a no-contact bin) and the final pair representation. Template inputs are supplied through RF3's token-level template track; template tokens are appended after query tokens in the modelled assembly, and query regions are sliced by component length for scoring.

4.2 Targets, structures, and MSAs

Thirteen two-state targets (Table 1) were curated from the fold-switching and conformational-diversity literature, with PDB accessions, constructs and chain choices verified against RCSB and PDBe SIFTS mappings. MSAs were built against UniRef30 (query plus up to 256 homologous rows; actual depths in Table 1). Templates for the conflict experiments were single-chain mmCIFs filtered from the corresponding state structures (chains kept under their original identifiers to avoid a silent template-dropping failure of the input pipeline; each template input was verified to produce a nonzero templated-atom count before inference). For GA88/GB88 the two state structures are the NMR models 2JWS and 2JWU [26]; the RBD template is the RBD chain of 6M0J [33].

4.3 Conformational-state preference score

For each target and condition, we compute the mean negative log-likelihood of the experimental Cα–Cα distances of each state under the predicted distogram: $\mathrm{NLL}_S = -\frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}} \log p_{ij}(d_{ij}^S)$, where $S \in \{A, B\}$ and $\mathcal{P}$ is the pair set. The state preference is pref $= \mathrm{NLL}_B - \mathrm{NLL}_A$ (positive favours state A). **Range matching:** each state's experimental map is aligned to the folded query sequence by sequence alignment (minimum matching block 5 residues), and $\mathcal{P}$ is restricted to residue pairs that are covered by *both* states' maps and finite in both, so that every preference compares identical residue pairs on both sides. This corrects the first-pass analysis, in which states were scored on their own spans (creating artificial preferences, e.g. LacY, EGFR) and a compressed disordered loop (UniProt 101–114 missing in the RfaH state-A reference) misaligned rows by 14 residues; all pre-correction magnitudes are superseded by the values reported here. The run-to-run noise band of pref shifts ($\pm$0.14–0.27) was estimated from archived repeat forwards of identical inputs. Steering and conflict measures are $\mathrm{steer}_{\mathrm{tpl}} = \mathrm{pref}_{\mathrm{tplB}} - \mathrm{pref}_{\mathrm{tplA}}$ and $\mathrm{conflict} = \mathrm{pref}_{\mathrm{msa,tplA}} - \mathrm{pref}_{\mathrm{tplA}}$.

4.4 Representation geometry and supervised evaluation on RBD

Pair representations were extracted for a 500-mutant subset of the SARS-CoV-2 RBD DMS dataset [31] under four input conditions: no-MSA, a 15-row MSA, the 6M0J RBD template, and MSA-plus-template. Geometry metrics per arm: effective rank (number of singular values needed for 90% of variance), linear CKA against the no-MSA arm, and per-mutant delta statistics (median cosine and median norm ratio of the mutant-minus-WT representation change relative to the no-MSA arm). Supervised evaluation trained a mutation-effect predictor on frozen representations with per-position standardization, using 5 cross-position splits (30% of positions held out) $\times$ 3 initializations, and reports best-epoch validation Spearman correlation; the de-biased full-set MSA effect quoted in the text (+0.14–0.15) comes from held-out-test and strict (train-only) standardization variants of the same pipeline on the full 3,998-mutant set.

4.5 Homodimer block analysis and the unpaired control

CASP3 was modelled as a two-chain, 488-token homodimer. For each of 392 mutants (stride-4 subset of the DMS set) we computed, per block (A–A, B–B, A–B) of the full pair representation, the cosine similarity and norm ratio between the MSA and no-MSA extractions, plus a paired Wilcoxon signed-rank test of A–B versus A–A cosines. The unpaired control re-ran the extraction with the chain-B MSA rows shuffled to destroy species-level pairing (100 mutants), using the same pipeline and statistics.

4.6 Reproducibility

Analysis scripts, target definitions, and all reported numbers are archived with the project; every figure is generated directly from archived JSON result files (`make_figs.py`). A claim-by-claim traceability table (`TRACEABILITY.md`) maps each quantitative statement to its source file.

Data availability

All structures used are public (PDB IDs in Table 1). DMS data are from published datasets [31, 32]. RF3 is publicly available [3]. Analysis code, inputs, and archived result files are publicly available at https://github.com/Lizon1c/rf3-state-selection.

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Competing interests

The author declares no competing interests.

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Table 1: Thirteen two-state targets and range-matched MSA-induced shifts. pref = $\mathrm{NLL}_B - \mathrm{NLL}_A$ (positive favours state A); shift = $\mathrm{pref}_{\mathrm{msa}} - \mathrm{pref}_{\mathrm{nomsa}}$. Direction is relative to the crystallized (PDB-dominant) state. MSA rows include the query. Noise band of the shift: ±0.14–0.27. XR = X-ray.

Target	Category	State A (PDB)	State B (PDB)	MSA rows	res.	$\mathrm{pref}_{\mathrm{no}}$	$\mathrm{pref}_{\mathrm{msa}}$	shift	direction
KaiB	fold-sw.	2QKE XR GS	5JYT NMR FS	257	90	+0.72	−1.57	−2.29	away
RfaH	fold-sw.	2OUG XR α	2LCL NMR β	257	42	+0.10	−2.89	−2.99	away
Mad2	fold-sw.	1DUJ NMR O	1S2H NMR C†	257	184	−0.05	−0.96	−0.91	toward
XCL1	fold-sw.	1J8I NMR chemok.	2JP1 NMR β-sand.	160	39	−0.78	+2.09	+2.87	toward
selecase	fold-sw.	4QHF XR mon.	4QHJ XR olig.	2	105	+0.13	+0.13	−0.01	null
GA88/GB88	designed	2JWS NMR 3α	2JWU NMR α/β	24	40	+0.52	+3.07	+2.56	sharpen‡
β2AR	GPCR	2RH1 XR inact.	3SN6 XR act.	257	271	−0.06	+0.09	+0.15	null
Abl1	kinase	2GQG XR DFG-in	2HYY XR DFG-out	257	262	−0.00	+0.01	+0.01	null
EGFR	kinase	1M17 XR αC-in	1XKK XR αC-out	257	265	+0.14	+0.12	−0.02	null
LacY	transporter	1PV6 XR inward	4OAA XR outward	257	388	+0.06	−0.01	−0.08	null
ubiquitin	XR/NMR	1UBQ XR	1D3Z NMR	257	76	−0.08	−0.10	−0.03	null
GB1	XR/NMR	1PGB XR	2GB1 NMR	89	55	+0.36	+0.44	+0.09	null
thioredoxin	XR/NMR	2TRX XR ox.	1XOA NMR ox.	257	108	+0.26	+0.38	+0.13	null

†For Mad2 the crystallized (PDB-dominant) state is state B (C-Mad2), so a negative shift points toward crystal.
‡GA88/GB88 has no single crystallized state; the shift sharpens each sequence's own-fold preference (main-flow GA88 scoring shown; per-sequence values in Fig. 3).

Table 2: Template versus MSA steering on five two-state targets (range-matched). Preferences as in Table 1; steer = $\mathrm{pref}_{\mathrm{tplB}} - \mathrm{pref}_{\mathrm{tplA}}$; conflict = $\mathrm{pref}_{\mathrm{msa,tplA}} - \mathrm{pref}_{\mathrm{tplA}}$. $|\mathrm{conflict}| \leq 0.23$ on all five: the template wins every direct conflict with the MSA.

Target	$\mathrm{pref}_{\mathrm{no}}$	$\mathrm{pref}_{\mathrm{msa}}$	$\mathrm{pref}_{\mathrm{tplA}}$	$\mathrm{pref}_{\mathrm{tplB}}$	$\mathrm{pref}_{\mathrm{msa,tplA}}$	steer	conflict
KaiB	+0.72	−1.57	+3.59	−3.00	+3.36	−6.59	−0.23
RfaH	+0.10	−2.89	+1.49	+0.30	+1.45	−1.19	−0.04
Mad2	−0.05	−0.96	−0.17†	−1.87	−0.17	−1.70	+0.00
XCL1	−0.78	+2.09	−1.49	−0.86	−1.44	+0.63	+0.05
LacY	+0.06	−0.01	+0.54	+0.08	+0.51	−0.47	−0.03

†Mad2 anomaly: the O-state template (1DUJ, NMR ensemble, N-terminal truncation) does not pull toward state A; see text.

Table 3: Template versus MSA on the RBD: geometry and function (500-mutant subset). eff90: effective rank (90% variance). CKA: linear CKA vs the no-MSA arm. Δdir./norm: median cosine and norm ratio of mutation-conditioned representation changes relative to no-MSA. Spearman: cross-position mutation-effect prediction (mean of 5 splits × 3 inits).

Arm	eff90	CKA vs no-MSA	Δ direction	Δ norm ratio	Spearman ρ
no MSA	181	(1.00)	(1.00)	(1.00)	0.415
MSA (15 rows)	139	0.568	0.48	1.21	0.638
template (6M0J)	160	0.443	0.26	1.51	0.366
MSA + template	159	0.448	0.26	1.52	0.370

CKA(template, MSA) = 0.482. The de-biased full-set MSA effect under the canonical protocol is +0.14–0.15 (Methods); the +0.223 here reflects the 500-subset protocol used for all four arms.